

Ashab Uddin

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PROFESSIONAL SUMMARY

AI Researcher (PhD) specializing in deep reinforcement learning, multi-agent systems, and AI-driven optimization for wireless edge and cyber-physical systems. Experienced in designing stable, constraint-aware decision-making frameworks for autonomous and real-time environments. Strong background in control systems, simulation, and mathematical modeling, with 10+ years of industrial automation experience bridging AI research with real-world deployment.

TECHNICAL SKILLS

AI & Optimization: Deep Reinforcement Learning, Multi-Agent Systems, Safe RL, Constraint Optimization (LP, NLP, MILP), Edge Intelligence, Wireless Resource Allocation, Distributed Systems, Control Systems, Stability Analysis

Frameworks: PyTorch, TensorFlow, Stable-Baselines3, AgileRL, Hugging Face

Programming: Python, C++, MATLAB

Wireless & Systems: Edge Computing, Vehicular Networks, Resource Allocation

Industrial Systems: PLC (Siemens, Allen-Bradley), VFDs, SCADA (WinCC, FactoryTalk)

Communication & Industrial Networks: Modbus TCP/IP, Industrial Ethernet, PROFINET, PROFIBUS, AS-i, PLC Networking, SCADA Communication, Real-Time Data Acquisition

Tools: Docker, Git, Linux

EXPERIENCE

Graduate Researcher

2022 – Present

University of Windsor

Windsor, Canada

- Proposed stability-aware multi-agent reinforcement learning (MARL) models with constraint-driven decision-making for safe and reliable operation in dynamic cyber-physical environments.
- Introduced Adaptive Reward-based Clustering and Sampling (ARCS) to improve experience replay efficiency, learning stability, and convergence in DRL systems.
- Designed simulation environments to stress-test AI agents under dynamic conditions, evaluating robustness, constraint violations, and performance reliability.
- Developed AI-driven optimization frameworks that improved system performance, efficiency, and reliability under latency, energy, and resource constraints.
- Published first-author research in IEEE venues including TVT, TCCN, TITS, VTC, and GLOBECOM.

Assistant Manager – Electrical & Automation

2018 – 2022

Transcom Beverages Ltd. (PepsiCo Franchisee)

Gazipur, Bangladesh

- Led optimization of industrial automation systems using PLCs and VFDs to improve production efficiency, reliability, and energy performance.
- Designed and implemented energy-efficient control strategies, reducing power consumption by 10% and improving production yield by 15%.
- Diagnosed and resolved real-time system failures in large-scale industrial control environments.
- Developed data-driven monitoring and control solutions to enhance system performance, operational stability, and reliability.
- Worked on large-scale automation systems including KRONES, KHS, and CO₂ (ASCO) plants.
- Established strong foundations in control and optimization, later extended to AI-driven decision-making frameworks.

Deputy Manager (Engineering)

2012 – 2017

Jeans 2000 Ltd.

Chattogram, Bangladesh

- Maintained and optimized DCS/SCADA-based systems (Siemens S7, B&R) for industrial processes including denim washing plant, reverse osmosis (RO) plant, and water treatment plant (WTP).
- Led system monitoring and control operations to ensure safety, reliability, and process efficiency.
- Conducted electrical and fire safety audits with international compliance bodies (Bureau Veritas, Accord, Alliance).
- Reduced energy consumption by 5% through VFD-based control optimization strategies.
- Improved energy efficiency by 15% using sensor-based intelligent control systems.

SELECTED INDUSTRY PROJECTS

Schneider ETAP-Based Energy Management System (EMS)

Implemented a Schneider ETAP-based EMS-SCADA system using Modbus TCP/IP communication to integrate field energy meters, PLCs, and plant-level monitoring systems for real-time energy monitoring, load analysis, and energy-efficiency improvement.

Installation and Commissioning of 33 kV/11 kV/0.4 kV, 4 MVA Substations

Led installation and commissioning of a 33 kV/11 kV/0.4 kV, 4 MVA two-stage substation, including medium-voltage distribution, transformer integration, protection coordination, and industrial electrical infrastructure setup.

Installation and Commissioning of 600 BPM KRONES CSD Beverage Production Line

Led installation and commissioning of a fully automated 600 BPM KRONES CSD beverage line, integrating PLC, VFD, SCADA, and Ethernet-based communication systems for real-time monitoring, synchronization, and high-speed production control.

SELECTED AI PROJECTS

DRL-based Resource Allocation in Vehicular Edge Computing

Designed reinforcement learning models optimizing latency, energy, and task completion under dynamic constraints with safety-aware decision policies.

Multi-Agent Reinforcement Learning for Distributed Control

Developed MARL frameworks (MADDPG, MATD3, COMA) enabling coordinated and stable decision-making under uncertainty in distributed systems.

Bilevel (Leader–Follower) Optimization Framework

Designed a single-leader multi-follower bilevel Stackelberg game-inspired DRL framework for coordinated resource allocation and distributed decision-making in dynamic edge computing environments.

LSTM-MHA-Assisted DRL Caching for Vehicular Edge Networks: San Francisco Cabspotting dataset

Contributed to an LSTM-MHA-assisted DDQN framework for proactive edge caching in vehicular networks, improving cache hit ratio, latency, and QoE through priority-aware caching decisions.

Traffic Sign Classification using Convolutional Neural Networks (CNNs): Kaggle's GTSRB dataset

Developed and optimized a CNN-based traffic sign classification system using Python and TensorFlow/Keras, performing comparative analysis of learning rates, batch sizes, network architectures, and optimizers (SGD vs Adam) to study convergence, overfitting, and classification performance.

SELECTED PUBLICATIONS

A. Uddin, A. H. Sakr, and N. Zhang, “Intelligent offloading in vehicular edge computing: A comprehensive review of deep reinforcement learning approaches and architectures,” arXiv:2502.06963, 2025.

A. Uddin, M. H. A. Shohag, A. H. Sakr, N. Zhang, and Q. M. J. Wu, “Adaptive Reward Clustering for Efficient Deep Reinforcement Learning in Vehicular Offloading,” *IEEE GLOBECOM*, 2025 (accepted).

A. Uddin, A. H. Sakr, and N. Zhang, “Counterfactual multi-agent DRL for efficient task offloading in vehicular edge computing,” *IEEE Transactions on Vehicular Technology*, 2025.

A. Uddin, A. H. Sakr, and N. Zhang, “Adaptive prioritization and task offloading in vehicular edge computing through deep reinforcement learning,” *IEEE Transactions on Vehicular Technology*, vol. 74, no. 3, pp. 5038–5052, 2024.

A. Uddin, A. H. Sakr, and N. Zhang, “Prioritized task offloading in vehicular edge computing using deep reinforcement learning,” *IEEE VTC-Spring*, 2024.

SELECTED COLLABORATIVE PUBLICATIONS

N. N. Pour, N. Zhang, and **A. Uddin**, “Proactive Content Caching in Urban Vehicular Networks for Connected Autonomous Vehicles,” accepted to IEEE INFOCOM 2026 Workshop on DOICT-IndSoc, 2026.

M. H. A. Shohag, Q. M. J. Wu, **A. Uddin**, and N. Zhang, “Bootstrapped Unseen Prototypes for Few-Shot Inspired Generative Zero-Shot Learning,” accepted to MiTA, 2026.

EDUCATION

University of Windsor

PhD in Electrical Engineering

Chittagong University of Engineering & Technology

BSc in Electrical & Electronics Engineering

Windsor, Canada

Expected Dec 2026

Bangladesh